BEN NGUYEN

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA — **December 2025**

Bachelor of Science in Computer Science

Concentration: Security and Privacy

Coursework: Applied Cryptography, Computer Security, Computer Networks, Database Systems, Distributed Systems, Artificial Intelligence, Software Engineering

EXPERIENCE

Software Quality Engineer – CACI – Herndon, Virginia — **January 2026 – Present**

- Developed Playwright automated testing framework for spectrum monitoring applications and integrated it into GitLab/Docker CI/CD pipelines, establishing scalable, repeatable quality assurance.
- Refactored legacy software interfaces with manufacturing equipment, improving maintainability of test processes.
- Identified and verified 50+ bugs across multiple defense product lines, ensuring high quality software delivery.

Artificial Intelligence Engineer Intern – FPT Software – Ho Chi Minh City, Vietnam — **May 2024 – August 2024**

- Improved online retail store's search engine performance by ~30% by evaluating embedding models using MAP@K values, Shapley values and LIME.
- Developed a large language model to create complex workflows from user inputs, significantly enhancing the UI/UX of a workflow creation application.

Teaching Assistant – Carnegie Mellon University – Pittsburgh, Pennsylvania — **January 2022 – December 2022**

- Led a review team providing feedback on assessments, assignments, class notes and other student-facing documents.
- Taught group sessions of 25 students, held office hours 2+ hours a week, tutored struggling students, created practice material, and graded assessments and assignments.
- Mentored 12 students through a 1000+ line project in Python showing algorithmic complexity and UI/UX skills.

PROJECTS

Hollow Knight Bot — **July 2026**

- Built an end-to-end RL system training a PPO agent against Hollow Knight, pairing a custom C# mod with a Python Gymnasium environment.
- Engineered full-stack fault tolerance, autonomous resets from cold boot, automatic recovery of crashed instances, and atomic checkpointing, enabling unattended overnight runs.
- Scaled to parallel game instances via copy-on-write app cloning for save isolation and asynchronous episode resets, cutting wall-clock training time by ~30% while keeping learning quality intact.

Overwatch Statistics Tracker — **June 2026 – July 2026**

- Built a full-stack web app for tracking Overwatch matches, using a Flask REST API with SQLAlchemy and a React + TypeScript dashboard that charts hero and map win rates.
- Added an AI scoreboard autofill via the Claude vision API, extracting stats from screenshots; validated with an eval harness and pytest suite.
- Designed a match-entry form that validates team composition, hero swaps, and bans in real time.

SKILLS

Languages: Python, C/C++, JavaScript, TypeScript, Rust, Standard ML, Golang, HTML/CSS, SQL

Tools: Git, Playwright, React, Node, LaTeX, AWS, Docker

Natural Languages: English, Vietnamese